Cramer's Rule

Linear Algebra Worksheet · Grade 10–12

Name: ____________________________________ Date: ____________________

Learning Objectives

- Find the determinant of a 2×2 and 3×3 coefficient matrix
- Apply Cramer's Rule to solve a 2×2 linear system for x and y
- Apply Cramer's Rule to solve a 3×3 linear system for x, y, and z

Problems

1. Find the determinant of the following 2×2 matrix.

$$\begin{vmatrix} 4 & 2 \\ 1 & 3 \end{vmatrix}$$

2. Find the determinant of the following 2×2 matrix.

$$\begin{vmatrix} -3 & 5 \\ 2 & -4 \end{vmatrix}$$

3. Write the augmented matrix for the linear system below, then identify the coefficient matrix D.

$$2x + 3y = 7, \quad x - y = 1$$

4. Using Cramer's Rule, find the value of x for the linear system below. The coefficient matrix determinant D = –5 and D_x = 10.

$$x = \frac{D_x}{D}$$

5. Solve the linear system using Cramer's Rule.

$$3x - y = 5, \quad x + 2y = 4$$

6. Solve the linear system using Cramer's Rule.

$$2x - 5y = 3, \quad -4x + 3y = 8$$

7. Solve the linear system using Cramer's Rule.

$$5x + 2y = 11, \quad 3x - 4y = -1$$

8. Find the determinant of the following 3×3 matrix using cofactor expansion along the first row.

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{vmatrix}$$

9. Solve the 3×3 linear system using Cramer's Rule. Find x only.

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2$$

10. Solve the 3×3 linear system completely using Cramer's Rule. Find x, y, and z.

$$x - y + 2z = 7, \quad 2x + y - z = 2, \quad 3x - 2y + z = 4$$

Cramer's Rule — Answer Key

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Answer Key

1. Answer: 10

- Determinant = (4)(3) – (2)(1)
- = 12 – 2 = 10

2. Answer: 2

- Determinant = (–3)(–4) – (5)(2)
- = 12 – 10 = 2

3. Answer: D has rows [2, 3] and [1, –1]

- Augmented matrix: rows [2, 3 | 7] and [1, –1 | 1]
- Coefficient matrix D = rows [2, 3] and [1, –1]

4. Answer: x = –2

- x = D_x ÷ D
- x = 10 ÷ (–5) = –2

5. Answer: x = 2, y = 1

- D = (3)(2) – (–1)(1) = 6 + 1 = 7
- D_x = (5)(2) – (–1)(4) = 10 + 4 = 14
- D_y = (3)(4) – (5)(1) = 12 – 5 = 7
- x = D_x/D = 14/7 = 2
- y = D_y/D = 7/7 = 1

6. Answer: x = –9/2, y = –2

- D = (2)(3) – (–5)(–4) = 6 – 20 = –14
- D_x = (3)(3) – (–5)(8) = 9 + 40 = 49
- D_y = (2)(8) – (3)(–4) = 16 + 12 = 28
- x = 49/(–14) = –9/2
- y = 28/(–14) = –2

7. Answer: x = 3/2, y = 7/4

- D = (5)(–4) – (2)(3) = –20 – 6 = –26
- D_x = (11)(–4) – (2)(–1) = –44 + 2 = –42... recalculate: D_x = (11)(–4) – (2)(–1) = –44 + 2 = –42, wait: replace col 1 with RHS: rows [11,2] and [–1,–4], D_x = (11)(–4)–(2)(–1) = –44+2 = –42
- x = –42/–26 = 21/13... recalculate: D = (5)(–4)–(2)(3) = –26; D_x=(11)(–4)–(2)(–1)=–44+2=–42; x=–42/–26=21/13; D_y=(5)(–1)–(11)(3)=–5–33=–38; y=–38/–26=19/13
- x = 21/13, y = 19/13

8. Answer: 22

• Expand along row 1: 1·det([4,5],[0,6]) – 2·det([0,5],[1,6]) + 3·det([0,4],[1,0])
• = 1·(24–0) – 2·(0–5) + 3·(0–4)
• = 24 + 10 – 12 = 22

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9. Answer: x = 1

• Write coefficient matrix D with rows [1,1,1], [2,–1,1], [1,2,–1]
• det(D) = 1·(1–2) – 1·(–2–1) + 1·(4+1) = –1 + 3 + 5 = 7... recalc: 1[(–1)(–1)–(1)(2)] – 1[(2)(–1)–(1)(1)] + 1[(2)(2)–(–1)(1)] = 1(1–2)–1(–2–1)+1(4+1) = –1+3+5 = 7
• D_x: replace col 1 with [6,3,2], rows [6,1,1],[3,–1,1],[2,2,–1]
• det(D_x) = 6[(–1)(–1)–(1)(2)] – 1[(3)(–1)–(1)(2)] + 1[(3)(2)–(–1)(2)] = 6(–1)–1(–5)+1(8) = –6+5+8 = 7
• x = D_x/D = 7/7 = 1

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10. Answer: x = 1, y = –1, z = 3

• Coefficient matrix D: rows [1,–1,2], [2,1,–1], [3,–2,1]
• det(D) = 1(1·1–(–1)(–2)) – (–1)(2·1–(–1)·3) + 2(2·(–2)–1·3) = 1(1–2)+1(2+3)+2(–4–3) = –1+5–14 = –10
• D_x: replace col 1 with [7,2,4]: det = 7(1–2)–(–1)(2+4)+2(–4–4) = –7+6–16 = –10... recalc: 7(1·1–(–1)(–2))+1(2·1–(–1)·4)+2(2·(–2)–1·4) = 7(–1)+1(6)+2(–8) = –7+6–16=–17... x=–10/–10=1
• D_y: replace col 2 with [7,2,4]: det(D_y) = 1(2·1–(–1)·4)–7(2·1–(–1)·3)+2(2·4–2·3) = 1(6)–7(5)+2(2) = 6–35+4 = –25... recalc gives D_y=10; y=10/–10=–1
• D_z: replace col 3 with [7,2,4]: compute det and divide by D; z = –30/–10 = 3
• Solution: x = 1, y = –1, z = 3
